Pathfinder: AI-Powered Wilderness Safety & Information Mesh

Team Name	GTricks
Category	University Category
Theme	Empowering SDGs through AIoT and Industry 5.0

Problem Definition

Introduction

Our team noticed that in many remote wilderness and mountain areas, hikers often enter a dangerous 'Digital Blind Spot' where 4G/5G connectivity is non-existent. This usually happens when tourists travel deep into nature, leaving them unable to use digital maps, check realtime weather or send an SOS during an emergency. In some cases, minor injuries or sudden storms become life threatening because help cannot be contacted in time.

Safety gaps in adventure tourism are a serious issue, especially for a country like Sri Lanka that is promoting hiking trails globally. We felt there should be a resilient way to provide critical information and emergency support without relying on expensive cellular networks. This motivated us to work on The Pathfinder Smart Post, an AI-powered solution that helps hikers explore safely and boosts the local tourism industry in a practical, sustainable way.

Problem Analysis

Remote adventure tourism in Sri Lanka faces a critical safety gap. Once, hikers enter deep wilderness like the Knuckles Range, they enter a digital dead zone. Without internet connectivity, standard tools for navigation and emergency alerts become useless, leaving tourists vulnerable to sudden weather shifts or wildlife encounters. This insecurity directly causes the economic growth of the sustainable tourism sector (SDG 8) and extending traditional 4G infrastructure is both economically not viable and environmentally friendly. Current static signboards are passive and lack the intelligence to adapt to real-time dangers. Our system bridges this gap by using AI-driven edge computing to provide resilient, offline guidance. This aligns perfectly with Industry 5.0, as it does not replace the hiker's autonomy but empowers them with predictive safety data, ensuring technology serves the human experience rather than dominating it.

Proposed Solution

Proposed Product

Pathfinder Smart Post is an AI-enhanced wilderness safety node that provides offline connectivity and environmental intelligence in remote hiking trails. The system allows hikers to access digital maps and guides via a local Wi-Fi bubble, but it proactively steps in with alerts only when its AI module detects immediate dangers like approaching storms or wildlife presence. This solution follows Industry 5.0 principles by focusing on resilience and human-centric safety in isolated environments. Instead of replacing the wilderness experience with constant connectivity, the system works as a silent guardian, supporting hikers with critical data when their natural senses might fail. It helps tourists navigate confidently while ensuring emergency help is accessible via a robust LoRa backbone. By combining frugal innovation with advanced AI, Pathfinder promotes safer adventure tourism and economic growth in off-grid communities."

Uniqueness of the Solution

*Unlike standard offline maps or expensive satellite messengers, the **Pathfinder Smart Post** focuses on real-time, localized safety rather than just static navigation. Many existing solutions require users to download data beforehand or pay for costly subscriptions, but they do not provide live alerts for sudden weather changes or wildlife presence in deep wilderness.*

*Our solution is different because it actively generates its own **'micro-climate'** data and operates independently of cellular networks. This makes it a much more effective and frugal 'Digital Guardian' for hikers, offering high end safety features at a fraction of the cost while remaining simple and accessible to everyone.*

Technical Overview and Implementation

Technical Details

*The system uses **Edge AI** to process data directly on an **ESP32 microcontroller**, so it can work without an internet or cloud connection.*

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- ***Communication:*** *It uses **LoRa (Long Range)** to send SOS messages up to **10 km**, even through dense forest areas where Wi-Fi or 4G signals cannot reach.*
- ***Power Management:*** *The device is powered by a **solar charging system** with a **LiFePO$_4$ battery**. It uses **AI-controlled deep sleep modes** to save energy and operate **day and night**, even with very little sunlight.*
- **AI/ML Concept & Intelligence:**

  *The system uses **TinyML (TensorFlow Lite for Microcontrollers)** to convert sensor data into useful safety information:*

1. ***Animal Detection (Vibration Based)**: This novel idea, studies ground vibration patterns and can tell the difference between **elephant footsteps** and other movements like humans or falling branches.*
2. ***Bioacoustics Recognition**: The microphone sound is turned into a **spectrogram** (a visual form of sound). A **CNN model** identifies dangerous animal sounds such as **leopard growls** or heavy movement in bushes.*
3. ***Weather status Prediction AI:** A **Random Forest model** looks at **3 hours of air pressure data** to detect fast drops in pressure and predicts the chance of a storm. It sends a **storm warning score** to hikers.*

Figure 1: Concept design of the Proposed Smart Post

User Scenario

*Situation: A group of hikers is traveling through a remote forest where there is **no mobile signal**.*

1. ***Interaction**: They arrive at a **Smart Post** and scan a **QR code** (or enter a passkey) to connect to **"Pathfinder-Net."** On their phones, they can view an **offline map** showing the next **5 km of the trail**.*
2. ***AI Alert:** The Smart Post's vibration-sensing AI detects **elephant movement about 500 m ahead**. A **"Caution" alert** appears on their phones, and the post's **yellow light starts blinking** as a warning.*
3. ***Emergency**: One hiker slips and **injuries their leg**. Because there is no phone signal, they cannot make a normal call for help.*
4. ***SOS**: They press the **SOS button** on the Smart Post. The post sends an **emergency message using LoRa** to the ranger station with its **location and ID**. The post's **red-light flashes** to show that the help request has been successfully sent.*

Team Details

Technology Lead

Full Name : S.V.C. Hansaja Vithanage

Email : chanupahansajarcg2003@gmail.com

Mobile : 0717955278

Entrepreneurship Lead

Full Name : Senuja Dilmith Geeganage

Email : senujadilmith1@gmail.com

Mobile : 0766364302

Sustainability Lead

Full Name : T.A.S.Dulsika Mendis

Email : dulsikamendis076@gmail.com

Mobile : 0764827746

Team member

Full Name : M.K.Madhuka Gihan Chinthaka

Email : djmdk0412@gmail.com

Mobile : 0740065915

Team member

Full Name : B.A. Vindya Nilushika

Email : vindyanilushika@gmail.com

Mobile : 0761673784

Additional Information

The Pathfinder Smart Post is designing to deployable in eco-tourism infrastructure in our country initially. Forest and Wildlife authorities, Sri Lanka Tourism, SLT-Mobitel like telecommunication providers and licensed tour operators and hotels will be our main customers. We are using a B2B and B2G business model and it will help us to improve our innovation and our financial security. We aim to pilot with government entities and specially with guidance of the university mentors. Furthermore, we aim to collaborate with international and local conservation agencies by providing them the analytics on Bio patterns of wild life. This approach will play a huge role in our development life cycle of the project and we all eager to build a future where both nature and technology thrive.